

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.2 REFLECTION AND CONGRUENCE: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why does a matatu's rear-view mirror show an image that is the same size and shape as the real object behind it, yet appears flipped, and how can we use this to prove that the original and its mirror image are always congruent? Remember the matatu conductor on the Nairobi–Thika highway holding a small rectangular mirror so the driver can see a lorry approaching from behind. Use everything you learned in Lessons 1 to 8 to answer each question below in the space under it. Write in full sentences, show all your working for calculations, and draw neat labelled diagrams with a ruler where asked. Use correct words such as mirror line, image, object, perpendicular bisector, congruent, SSS and lateral inversion.

Part 1: What the mirror does to the lorry

On the Nairobi–Thika highway the conductor's mirror shows three things about the lorry's image: (a) it appears as far behind the mirror surface as the real lorry is in front of it, (b) it is the same size as the lorry, and (c) the lorry's right headlight appears on the left. Draw a labelled diagram showing the lorry (object), the mirror line and the image. Then use the properties of reflection you discovered in Lesson 4 to explain each of the three observations (a), (b) and (c).

Write your answer here:

Part 2: Coordinates and the mirror line

(a) Imagine the matatu mirror is the y-axis of a Cartesian plane. Write the rule that maps any object point (a, b) to its image, and explain in one sentence what stays the same and what changes. **(b)** A point $P(2, 1)$ has the image $P'(6, 5)$ after reflection in an unknown mirror line. Find the equation of the mirror line. Show each step of your working and say which facts about the mirror line you used.

Write your answer here:

Part 3: Proving the image is congruent

Triangle NMK is reflected in a mirror line to give triangle N'M'K'. (a) Explain why the sides NM, MK and KN are equal in length to N'M', M'K' and K'N'. (b) Name the congruence test that proves triangle NMK is congruent to triangle N'M'K', and write the congruence statement. (c) Why do we not need to measure all six elements (three sides and three angles) of the two triangles?

Write your answer here:

Part 4: Why the image looks flipped

A classmate says: 'If the lorry's image is congruent to the lorry, it should be possible to place one exactly on top of the other without lifting it off the page, so the right headlight should not appear on the left.' Respond to your classmate. In your answer, name the type of congruence that a reflection produces and explain how it resolves the puzzle in the driving question.

Write your answer here:

Part 5: Using reflection and congruence in real life

(a) Give two examples from Kenyan daily life where reflection is used or can be seen, and one example where congruence is used or can be seen. (b) Look back at the model you drew in Lesson 1. Describe two ideas in it that you would keep and two things you would change or add now, using at least two terms from this unit.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the phenomenon with properties of reflection	Correctly explains equal distance, equal size and lateral inversion, linking each to the matatu mirror and the lorry, with a fully labelled diagram showing perpendicular connectors.	Explains at least two of the three observations correctly using the properties of reflection; the diagram is mostly labelled.	Describes what is seen but does not link it to properties of reflection, or the diagram is missing or unclear.
Coordinate rules and mirror-line equation	States (a, b) to (-a, b) correctly and finds $y = -x + 7$ with all steps shown (midpoint, gradient, negative reciprocal, equation) and a reason for each.	Finds the correct equation with minor slips in working or missing reasons, or states the rule correctly with a small error in the equation.	Cannot state the rule or find the midpoint and gradient correctly, and does not reach the equation.
Congruence reasoning (SSS and opposite congruence)	Explains that reflection preserves distances, names SSS with a correct congruence statement, and explains why six elements are not needed and why the congruence is opposite.	Names SSS and gives a correct statement but the explanation of why distances are preserved or why the congruence is opposite is incomplete.	Does not name a valid congruence test or confuses congruence with orientation.

Resolving the driving question and using Kenyan contexts	Clearly distinguishes size/shape from orientation, answers the driving question in a connected argument, and gives accurate Kenyan examples and a thoughtful model revision.	Answers the driving question with some links between ideas; examples and model revision are accurate but brief.	Gives separate, unconnected statements; examples are missing or inaccurate and there is little model revision.
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